EXPERIMENT 17

High Level Programming Extensions – Procedures

Aim

To create and execute a MySQL stored procedure to display employee details for a given department.

Software Required

MySQL Server and MySQL Workbench or MySQL command-line client.

SQL Code

```
CREATE DATABASE IF NOT EXISTS CompanyProcedureDB;
USE CompanyProcedureDB;

DROP TABLE IF EXISTS Employee;
CREATE TABLE Employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    department VARCHAR(30),
    salary DECIMAL(10,2)
);

INSERT INTO Employee VALUES
(101, 'Ravi', 'IT', 45000.00),
(102, 'Priya', 'HR', 38000.00),
(103, 'Arun', 'IT', 52000.00),
(104, 'Meena', 'Sales', 41000.00);

DROP PROCEDURE IF EXISTS GetEmployeesByDepartment;
DELIMITER //
CREATE PROCEDURE GetEmployeesByDepartment(IN dept_name VARCHAR(30))
BEGIN
    SELECT emp_id, emp_name, department, salary
    FROM Employee
    WHERE department = dept_name;
END //
DELIMITER ;

CALL GetEmployeesByDepartment('IT');
```

Expected Output:

```
mysql> CREATE DATABASE company_db;
Query OK, 1 row affected (0.05 sec)

mysql> USE company_db;
Database changed
mysql> CREATE TABLE employee (
    ->     emp_id INT PRIMARY KEY,
    ->     emp_name VARCHAR(50),
    ->     salary DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO employee VALUES (101, 'Ravi', 25000);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO employee VALUES (102, 'Priya', 35000);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO employee VALUES (103, 'Kiran', 45000);
Query OK, 1 row affected (0.04 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION annual_salary(salary DECIMAL(10,2))
    -> RETURNS DECIMAL(12,2)
    -> DETERMINISTIC
    -> BEGIN
    ->     RETURN salary * 12;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT annual_salary(25000) AS Annual_Salary;
+---------------+
| Annual_Salary |
+---------------+
|     300000.00 |
+---------------+
1 row in set (0.04 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION calculate_bonus(salary DECIMAL(10,2))
    -> RETURNS DECIMAL(10,2)
    -> DETERMINISTIC
    -> BEGIN
```




```
mysql>
mysql> DELIMITER ;
mysql> SELECT annual_salary(25000) AS Annual_Salary;
+---------------+
| Annual_Salary |
+---------------+
|     300000.00 |
+---------------+
1 row in set (0.04 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION calculate_bonus(salary DECIMAL(10,2))
    -> RETURNS DECIMAL(10,2)
    -> DETERMINISTIC
    -> BEGIN
    ->     RETURN salary * 0.10;
    -> END //
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT emp_id, emp_name, salary,
    ->        annual_salary(salary) AS annual_salary,
    ->        calculate_bonus(salary) AS bonus
    -> FROM employee;
+--------+----------+----------+---------------+---------+
| emp_id | emp_name | salary   | annual_salary | bonus   |
+--------+----------+----------+---------------+---------+
|    101 | Ravi     | 25000.00 |     300000.00 | 2500.00 |
|    102 | Priya    | 35000.00 |     420000.00 | 3500.00 |
|    103 | Kiran    | 45000.00 |     540000.00 | 4500.00 |
+--------+----------+----------+---------------+---------+
3 rows in set (0.00 sec)

mysql> DROP FUNCTION IF EXISTS annual_salary;
Query OK, 0 rows affected (0.01 sec)

mysql> DROP FUNCTION IF EXISTS calculate_bonus;
Query OK, 0 rows affected (0.03 sec)

mysql>
```



Observation

The stored procedure accepts a department name as an input parameter and displays matching employee records.

Result

Thus, a MySQL stored procedure was created and executed successfully.